

Cyber Security Assignment Brief

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Module Details			
Module code:	COM7038M-2025-26	Level of Study:	7
Module Leader(s):	Dr Soonleh Ling (Module Director) Ms Niharika Gottipatti (Deputy Module Director)	Credits:	15
Assessment format:	Lab Report (Anonymous)	Method of submission:	Turnitin within Moodle
Deadline or Assessment Period:	9 th June 2026, 12Noon	Feedback date and place:	1 st July 2026, Written feedback within Turnitin/Moodle
Assessment limits: length, load, word count, etc.	N/A	Component number:	1 of 1
Is this exempt from anonymous marking under the policy?	No	Component weighting:	100%

Assignment Description

This coursework is designed to demonstrate the broad understanding and knowledge of the Cyber Security, assessing, and evaluating the student's strength and level of analysis. The assessment consists of a series of questions, with the objectives to achieve the learning outcomes from the module.

You are required to produce an independent report to critically evaluate the questions, paying particular attention to the level of depth required to answer those questions by applying the learning in both online lectures and seminar sessions.

It is important that the content is underpinned with the inclusion of relevant academic theory, concepts, and models where appropriate, as well as contemporary industrial insights. These should be accurately cited and referenced according to [York St John Harvard Referencing](#) throughout.

Please find the detailed assessment criteria for each part in the Assessment Marking Information section.

Part 1 – [20 marks]

1. Penetration testers use footprinting to build a comprehensive profile of a target:
 - a. Explain the concept and importance of Footprinting.
 - b. Discuss both technical and non-technical methods used in Footprinting.
 - c. Describe measures that organisations can implement to protect sensitive data and defend against unauthorised or malicious Footprinting attempts.
2. Explain the methods of DNS Enumeration and discuss two standard tools that may be used during these activities with practical examples.

Part 2 – [25 marks]

1. What is the Common Vulnerabilities and Exposures (CVE) system?
 - What are CVE Identifiers?
 - When are you likely to need to refer to CVE Identifiers during a security test?
2. Name a well-known automated vulnerability scanner. Discuss the merits of performing vulnerability identification:
 - Manually
 - using automated vulnerability scanners
3. Assume that during the assessment stage, you have compromised a host machine in the target network under a normal user account. Discuss two techniques you may attempt to obtain higher privileges in the compromised host.
4. Discuss the testing method known as Fuzzing and how it can be used during the security assessment of an organisation.

Part 3 – [25 marks]

1. Describe the method of port scanning with an example and explain the difference between open, closed, and filtered port states.
2. In a scenario when port scanning discovers open ports on a target system, how can encryption be implemented to prevent unauthorised access?
3. Explain two common encryption algorithms used in modern cybersecurity practices? Compare their cryptographic features and strengths in protecting sensitive data.

Part 4 – [20 marks]

1. Explain the different steps of ethical hacking and provide the aim of each step.
2. Give two real-world examples of ethical challenges that ethical hackers face during these steps.

Learning Outcomes (LOs)

Programme Learning Outcomes (PLOs) 7.1-7.7:

- 7.1 Evaluate computer science concepts and principles and their application to the effective design, implementation, and usability of computer-based systems.
- 7.2 Apply the findings of advanced scholarship and/or contemporary research and practice to the solution of computer science problems
- 7.3 Critically evaluate computer science problems, including those at the forefront of the field.
- 7.4 Demonstrate operation within applicable professional, legal, social, and ethical frameworks.
- 7.5 Demonstrate originality and creativity in the solution of computer science problems.
- 7.6 Recommend, with detailed justification, the appropriate computer science principles, and practices to apply to the significant domain-specific activity.
- 7.7 Apply standards, quality processes, and engineering principles to the solution of computer science problems.

Advice and Guidance

Guidelines for Students:

Students must submit their own work. They must acknowledge the sources used in this assignment, failure to acknowledge would be plagiarism which is an academic offence and a penalty can be imposed. Students need to write by reading other papers on their own with citations and leave references at the end of the assignment.

Students work would be submitted to the national plagiarism facility. This identifies the sources from the internet and other extensive databases. Once the student's work has been submitted to detection services, work is stored in databases electronically and compared their work from other sources. It is necessary to keep a backup of their work. Students' materials would be stored in the database electronically for indefinite periods.

It is essential that you acknowledge the source of any research, information, ideas, opinions, theories, or other material which is not your own. Effective referencing, quoting, paraphrasing, and summarising show evidence of the reading you have done and ensure that you avoid accusations of plagiarism.

The University's fundamental stance on the use of Turnitin is geared toward supporting students' academic development. You can use this link to check your work for areas where you might be at risk of plagiarising.

Please submit your assignment on time. All assignments may be electronically submitted using Turnitin (via Moodle) by midnight on the due date. Please do not submit your assignment last minute. Please also allow time for any problems or issues with systems.

The work you present should be your own work, and not just copied from others. You can quote from others, but you must say who the author is and use quotation marks or paraphrase. If you do not do so, we will investigate your work for academic misconduct. This is particularly likely if your Turnitin similarity score is high and/or individual matches are high.

If you require support with your study skills, please visit <https://www.yorks.ac.uk/students/study-skills/> or alternatively, you can also visit Student Hub [Home - Student Hub Online Portal](#)

Please refer to the York St John University Code of Practice for Assessment and Academic Related Matters 2023-24.

We ask that you pay particular attention to the academic misconduct policy. Penalties will be applied where a student is found guilty of academic and/or ethical misconduct, including termination of programme ([Policy Link](#)).

You are also required to keep to the word limit set for an assessment and to note that you may be subject to penalty if you exceed that limit. You are required to provide an accurate word count on the cover sheet for each piece of work you submit ([Policy Link](#)).

For late or non-submission of work by the published deadline or an approved extended deadline, a mark of 0NS will be recorded. Where a re-assessment opportunity exists, a student will normally be permitted only one attempt to be re-assessed for a capped mark ([Policy Link](#)).

An extension to the published deadline may be granted to an individual student if they meet the eligibility criteria of the ([Policy Link](#)).

Additional Information

The report should be created using Microsoft Word. You have access to Microsoft software for free through your university account.

Your report should include the following areas:

1. Student ID number, name, date, title of the assessment
2. Assignment questions
3. Harvard reference list

The report should contain all the relevant information but also be professional including the use of graphs, tables, and figures as appropriate. You will be assessed on the creativity of your report, application of theory and knowledge, as well as your use of research and referencing throughout.

Your work needs to:

- Be underpinned by a range of contemporary academic and industrial research.
- Critically appraise the literature and theories relating to the issue under review; demonstrate awareness and the application of theory to practice.
- Provide practical examples to demonstrate awareness and the application of theory to practice.

Assessment Marking Information

Your work will be marked according to the assessment instructions provided within this document and the selected Learning Outcomes' (LOs) (see above).

This assessment is marked using the assessment marking criteria or a similar rubric that aligns with the University's Generic Assessment Descriptors (GAD).¹ This is to ensure all assessment decisions are comparable regardless of the discipline or mode of assessment.

Please note that you **must** meet the required baseline standards (50 – 59%) which will include the LOs and minimum expectations of the assessment. Further still, you must ensure you meet the requirements of each grade boundary to progress to the next, i.e., you should demonstrate your learning through the standards of the Pass, Merit and Distinction to reach a required Distinction (70 – 84%). These standards are designed to scaffold and build your learning to achieve your fullest potential in each criterion being assessed.

¹ A rubric is a type of scoring guide that markers use to set out specific components and expectations for an assignment for their students. It is then used to guide the marking they undertake.

Question Part	Marking Criteria	Marks
Part 1	1. Learner should demonstrate comprehensive subject knowledge on the Footprinting stage of ethical hacking. ○ Learner should be able to explain the concept and importance of Footprinting. (2 marks) ○ Learner should discuss both technical and non-technical methods used in Footprinting. (4 marks) ○ Learner should describe measures that organisations can implement to protect sensitive data and defend against Footprinting attempts (4 marks) 2. Learner should explain the methods of DNS Enumeration and should also discuss two standard tools that may be used with DNS enumeration with practical examples. (10 marks)	20
Part 2	1. Learner should demonstrate comprehensive subject knowledge and conceptual understanding of Cyber Security CVE system, particularly the CVE identifiers and its usage in the security testing. ○ Learner should state on what are CVE Identifiers. (3 marks) ○ Learner needs to provide a reason on when we need to refer to CVE Identifiers during a security test. (3 marks) 2. Learner should name a well-known automated vulnerability scanner and should discuss the merits of performing vulnerability identification: ○ Manually and (3 marks) ○ using automated vulnerability scanners (3 marks) 3. Learner should discuss two techniques that can be used to obtain higher privileges in the compromised host. (6 marks)	25

	4. Learner should discuss the testing method known as Fuzzing and how it can be used during the security assessment of an organisation. (7 marks)	
Part 3	1. Learner should demonstrate comprehensive subject knowledge and conceptual understanding of Cyber Security port scanning with example and explain the differences of three possible port scanning states i.e., open, closed and filtered. (10 marks) 2. Learner should show independent thinking of a scenario when port scanning discovers open ports on a target system, how can encryption be implemented to prevent unauthorised access? (5 marks) 3. Learner should explain two common encryption algorithms used in modern cybersecurity practices. Compare their cryptographic features and strengths in protecting sensitive data. (10 marks)	25
Part 4	1. Learner should demonstrate comprehensive subject knowledge and conceptual understanding of Cyber Security ethical hacking, information about ethical hacking steps, and in particular information about the aims and objectives of each step by describing the different steps of ethical hacking. (10 marks) 2. Learner should demonstrate in his/her answer the critical and independent thinking, able to understand fully the ethical challenges that ethical hackers face during each stage of ethical hacking with two examples. (10 marks)	20

Communication	1. Learner should present his/her answer clearly and in good logical manner. 2. Learner should provide any relevant references used in the answers, diagrams, figures, tables, concepts, and ideas. 3. Learner should provide any relevant references in his/her answers and adhere to YSJU's academic conventions such as Harvard referencing and plagiarism. 4. Learner should present his/her answers in clear written English, free from spelling errors, typo errors, and professional report format.	10
Total Marks		100

Detailed Marking Criteria

Total Available Marks (%): 100

Pass Grade Bands (100 – 50) (Learning Outcomes must be met)

Fail Grade Bands (49 – 0) (Learning Outcomes are not met)

Note that the detailed marking criteria below are derived from the YSJU GAD table.

Criteria	Pass (50–59)	Merit (60–69)	Distinction (70–84)	Distinction (85–100)	Borderline Fail (45–49)	Fail (30–44)	Fail (0–29)
Part 1: Footprinting and DNS Enumeration	The learner provides a satisfactory explanation of the concept of Footprinting and DNS Enumeration, identifying key methods and tools with some	The learner provides a good analysis of Footprinting and DNS Enumeration, with clear examples of technical and non-technical methods and an	The learner provides an excellent, detailed explanation of Footprinting, DNS Enumeration, and defence measures. The examples are well-reasoned and supported with practical insights.	The learner demonstrates exemplary understanding, providing new insights into Footprinting, DNS Enumeration tools, and protection methods. The	The learner's explanation of Footprinting and DNS Enumeration lacks depth, with limited discussion of defensive measures.	The learner shows minimal understanding of Footprinting and DNS Enumeration. Explanations are incomplete or incorrect.	The learner fails to explain Footprinting or DNS Enumeration. Little or no relevant content is provided.

	understanding of defensive measures.	explanation of defence strategies.		explanations show advanced thinking and application of concepts.			
Part 2: CVE System, Vulnerability Scanning, and Fuzzing	The learner provides a satisfactory explanation of the CVE system and vulnerability scanning, identifying automated scanners with a basic understanding of manual vs. automated approaches. Fuzzing is briefly explained, with some relevance to security assessments.	The learner demonstrates a good understanding of the CVE system and explains the merits of manual and automated vulnerability scanning with clear examples. Fuzzing is explained with practical relevance to the security assessment process.	The learner offers an excellent, detailed analysis of CVE Identifiers, vulnerability scanning methods, and Fuzzing, with insightful comparisons between manual and automated approaches. Fuzzing is explained thoroughly, highlighting its significance in security assessments.	The learner demonstrates exemplary understanding, offering new perspectives on CVE Identifiers, vulnerability scanners, privilege escalation techniques, and Fuzzing. The explanations are highly detailed, and Fuzzing is discussed as an advanced tool in security assessment.	The learner's explanation of the CVE system, vulnerability scanning, and Fuzzing is incomplete, with limited analysis or relevance of Fuzzing in security assessments.	The learner shows minimal understanding of the CVE system, vulnerability scanning, or Fuzzing, with unclear or incorrect explanations.	The learner fails to explain the CVE system, vulnerability scanning methods, or Fuzzing. Little or no relevant content is provided.
Part 3: Port Scanning and Encryption	The learner provides a basic explanation of port scanning, identifying key scanning states and encryption algorithms with some understanding of their cryptographic features.	The learner demonstrates a good understanding of port scanning and encryption, providing clear explanations of scanning states and cryptographic features with relevant examples.	The learner offers excellent, detailed explanations of port scanning, encryption, and cryptographic features. The comparisons are insightful and well supported with examples.	The learner demonstrates exemplary understanding of port scanning and encryption, providing innovative insights into cryptographic algorithms and port scanning methods. The explanations are highly detailed and professionally reasoned.	The learner's explanation of port scanning and encryption lacks clarity, with limited discussion of cryptographic features or scanning states.	The learner shows minimal understanding of port scanning or encryption, with unclear or incorrect explanations.	The learner fails to explain port scanning or encryption algorithms. Little or no relevant content is provided.
Part 4: Ethical Hacking	The learner provides a basic explanation	The learner demonstrates a	The learner offers an excellent, detailed	The learner demonstrates	The learner's explanation of	The learner shows minimal	The learner fails to explain the

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	of the steps of ethical hacking, identifying key objectives and challenges with some understanding of ethical considerations.	good understanding of the ethical hacking process and ethical challenges, providing clear examples and explanations of each step.	analysis of ethical hacking steps and challenges, with insightful explanations and examples of ethical considerations.	exemplary understanding of ethical hacking, providing innovative insights into the process and ethical challenges. The explanations are highly detailed and demonstrate advanced thinking.	ethical hacking lacks depth, with limited discussion of challenges or ethical considerations.	understanding of ethical hacking, with unclear or incorrect explanations.	steps or ethical challenges of ethical hacking. Little or no relevant content is provided.
Communication	The learner presents their work clearly, adhering to some academic conventions, with minor errors in referencing or structure.	The learner produces well-written and logically organized work, following academic conventions with good adherence to YSJU's standards.	The learner's work is excellent, clearly written, well-referenced, and free from errors. The learner adheres to academic conventions throughout.	The learner's work is exemplary in clarity, organization, and adherence to academic standards. References are consistently applied, and the writing is error-free.	The learner's presentation is unclear, with notable issues in referencing and structure. Some academic conventions have been overlooked.	The learner's work has major issues with clarity and referencing. Significant academic conventions are not followed.	The learner fails to adhere to basic academic conventions, with little or no effort to present clear writing or follow proper referencing guidelines.